

Past Papers

2014	2015	2016	2017	2018
------	------	------	------	------

Q Explain RACI charts?

(2014), (2017)

Ans:- Some organizations use ~~also~~ RACI charts to show four key roles for Project Stakeholders:

Responsible: Who does the task?

Accountable: Who has authority for it?

Consulted: Who has information necessary to complete the task?

Informed: Who needs to be notified of task status and results?

Q Explain Staffing Management Plan?

(2014)

Ans:- A Staffing management plan describes when and ~~with~~ how people will be added to the project team and taken off it. The level of detailed may vary based on the type of project.

e.g:- If an IT project is expected to need 100 people per year, the staffing management plan would describe the types of people needed to work on the project, such as Java Programmers, business analysts, technical writers and the number of each type of ^{person} ~~people~~ needed each month.

Q Explain Risk Register?
(2014)

Ans:- A Risk Register is a document that contains results of various risk management processes; it is often displayed in a table or spreadsheet format. A Risk Register is a tool for documenting potential risk events and related information. Risk events refer to specific, uncertain events that may occur to detriment or enhancement of the project.

Q Differentiate "Project" and "Operation"?
(2014)

Ans:- Project

Operation

→ Projects are unique and temporary.

→ Project have a fixed budget.

→ Project create a unique product, service or result

→ Construct a ~~new~~ Building, Design a car are the example of Project.

→ Operations are on-going & permanent with repetitive output

→ Operations have to earn profit in order to run the business.

→ Operations produce the same product, aim to earn profit, and keep the system running.

→ Any manufacturing or production process are the example of operation.

Q Name the three techniques of Project Cost Estimation / Budget Estimation?
(2014), (2015), (2017)

Ans:- Three techniques of Cost Estimation are:-

- 1) Top-Down Estimates
- 2) Bottom-Up Estimates
- 3) Parametric Estimates.

Q If there are five team members, how many communication channels will be there?
(2014)

Ans:- Number of Communication Channels = $n(n-1)/2$
 $= 5(5-1)/2$
 $= 5(4)/2$
 $= 20/2$
 $= 10$ Ans

Q If there is 12 team members, how many communication channels will be there?
(2016)

Ans:- Number of Communication Channel = $n(n-1)/2$
 $= 12(12-1)/2$
 $= 12(11)/2$
 $= 132/2$
 $= 66$ Ans

Q Differentiate "Project Manager" and "Sponsor"?
(2014)

Ans:- Project Manager:-

A Person who provides leadership and direction for the project managers heading the projects within a program.

Sponsor:-

A person who provides the direction and funding of the project is called sponsor.

Q Name the three techniques of Conflict Resolution?
(2014)

Ans:- The techniques of Conflict Resolution are:-

- Competing
- Collaborating
- Accommodating
- Compromising
- Avoiding

Q Name & Explain shortly triple Constraints?
(2015)

Ans:- The Triple Constraints are:-

Cost: The financial constraints of a project, also known as the project budget.

Scope: The tasks Required to fulfill the project's goals.

Time: The schedule for the project to reach completion.

The triple constraints states that the success of the project is impacted by its budget, deadlines and features.

or?

Q Differentiate "Direct" and "Indirect" Cost with Example?
(2015)

Ans:- Direct

Indirect

program.

and

solution?

n are:-

→ Direct cost are the costs that can be directly related to producing the products and services of the project.

e.g:- Salaries, cost of hardware & Software etc

→ Indirect cost are the costs that are not directly related to products or services of the project, but are indirectly related to performing the project.

e.g:- Cost of Electricity, Paper towels etc.

Q Define Benchmarking? or List reasons why organizations use benchmarking?
(2015), (2017)

ts?

t,

ject's

s of
ndlines

Ans:- Benchmarking defines "A Measurement of the quality of an organization's policies, products, programs etc and their comparison with standardised measurements of its peers."

The Objectives/Reasons of benchmarking are:-

- 1) To determine what and where improvements are called for.
- 2) To analyze how other organizations achieve their performance levels.
- 3) To use this information to improve performance.

Q Differentiate "Program" & "Project"?

(2015), (2018)

Ans:- Project

Program

→ A Project is a temporary endeavor undertaken to create a unique product or service.

→ Project focus on content.

→ It is short-term

→ It produces output

→ The functional units of project is single.

→ It has narrow scope

→ A program is a group of related projects in a coordinated way to obtain benefits.

→ Program focus on context

→ It is long-term

→ It produces outcome

→ The functional units of program is multiple.

→ It has broader scope

Q Describe two types of software that can assist in Project Quality Management?

(2016)

Ans:- These are:-

- 1) Spreadsheet & Charting Software: helps to create Pareto diagrams, Fishbone diagrams etc.
- 2) Statistical software Packages: helps to perform statistical analysis
- 3) Specialized Software Products: helps to manage six sigma projects or create quality control charts
- 4) Project management software: helps to create Gantt charts.

Q Briefly explain five Basic techniques (modes for handling conflicts) of conflict Resolution?
(2016)

Ans:- These are:-

- 1) Competing: Such as powering your way to win or defending your position.
- 2) Accomodating: by sub-ordinating your own interests to the interests of others.
- 3) Avoiding: by denying the existence of the conflict or withdrawing from it.
- 4) Callaborating: engaging and working together toward a solution
- 5) Compromising: Aggs Agree on a partially acceptable solution

Q How to Acquire the project team?
(2016)

Ans:- Acquiring the project team means assigning the appropriate staff to work on the project.

Acquiring the project team involves:-

- Knowing pre-assigned resources
- Negotiating for best possible resources
- Preventing failure to acquire the right resources
- Hiring from outside the organization
- Virtual teams
- Issues and Benefits
- Managing Risks associated with Resources.

Q Explain RAM?

(2016)

Ans:- RAM stands for Responsibility Assignment Matrix. It is also known as Responsibility Accountability matrix or RACI matrix is a key component of completing a project successfully. It is a matrix that maps the work of a project, as described in WBS and Responsible for performing the work, as describe in the OBS (Organizational Breakdown Structure).

Q Explain Projectized Organization?

(2017)

Ans:- An organizational in which the project manager has fully authority to assign priorities, apply resources and direct the work of persons assign to the project is called projectized organization. In projectized organization, there is clear line of authority and fast decision making.

Q Enlist the processes involved in project communication management?

(2017)

Ans:- The processes involved in PCM are:-

- Identify Stakeholders
- Plan Communication
- Distribute Information
- Manage Stakeholders expectations
- Report Performance.

Q Define Project Baseline?

(2017)

Ans:- A Project baseline is the original and approved project plan and depicts all key information and constraints. It includes scope baseline, cost baseline and schedule baseline. It allows the project team to measure their performance against the expectations and requirements that were set.

Q Explain Project time Management processes in detail?

Ans:- Project time Management:-

A Subset of Project management that includes: "The processes required to manage timely completion of the project".

Project Time Management Processes:-

The Project Time Management Processes consist of six steps. These are:-

1) Activity Definition:-

Activity definition involves developing a more detailed WBS and supporting explanations to understand all the work to be done, so you can develop realistic cost and duration estimates. In this step, we can

identifying the specific activities that the project team members and stakeholders must perform to produce a project deliverables.

2) Activity Sequencing:-

It identifying and determining the relationship between project activities. It involves reviewing activities and determining dependencies. You must use determine dependencies in order to critical path analysis.

3) Activity Resource Estimating:-

Before Estimating activity durations, you must have a good idea of the quantity and type of resources that will be assigned to each activity. In this step, we estimating how many resources a project team should use to perform project activities.

4) Activity Duration Estimating:-

Estimating the number of work periods that are needed to complete individual activities. Duration includes the actual amount of time worked on an activity plus the elapsed time. And Effort is the number of workdays or work hours required to complete a task.

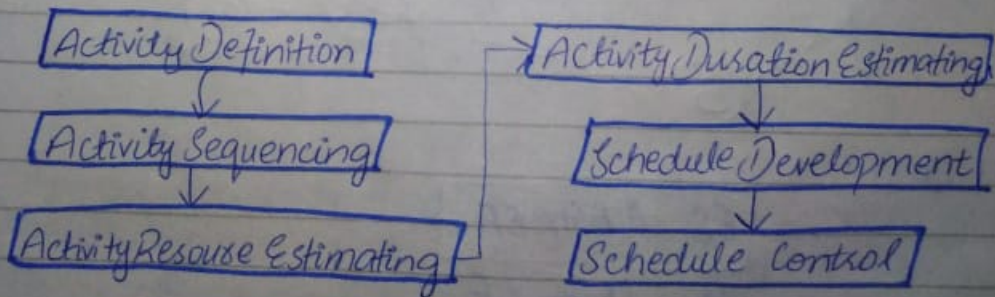
5) Schedule Development :-

It can analyzing activity sequences, activity resource estimates and activity duration estimates to create the project schedule. that provides a basis for monitoring project progress for the time dimension of the project. Important tools and techniques include Gantt Charts, critical path analysis, PERT analysis etc.

6) Schedule Control:-

It can controlling and managing changes to the project schedule. This process explains project status monitoring, finding changes in the schedule and control schedules changes. Communication is more important in this process. Tools and techniques includes progress reports, Variance analysis, A schedule change control system, performance management etc.

Diagram of PTMP:-



Q a) Draw a network diagram for following table?

b) Calculate Early Start (ES), Early Finish (EF), Late Start (LS), Late Finish (LF) and Total Slack for each activity?

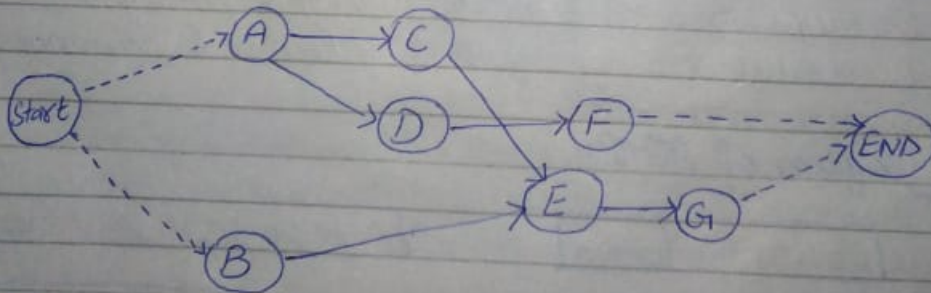
c) Show and explain Critical Path?

(2014), (2016)

Activity	Time	Predecessors
A	10	-
B	7	-
C	5	A
D	13	A
E	4	B, C
F	12	D
G	14	E

Ans:-

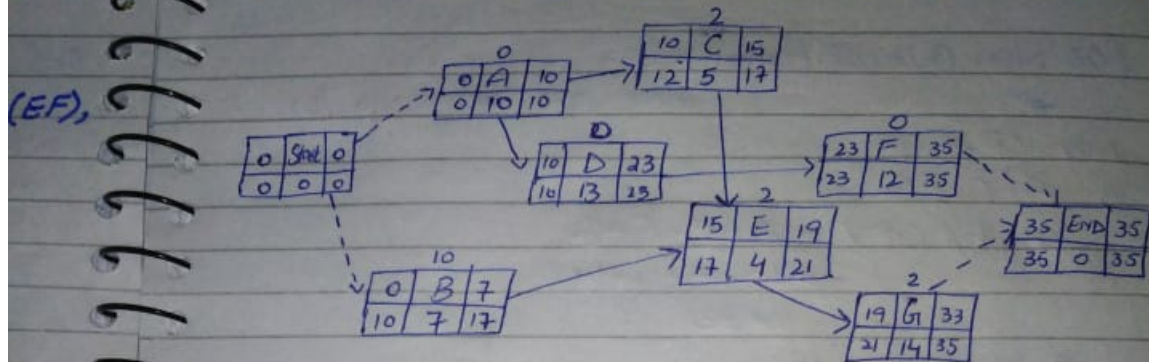
a) Network Diagram:-



b)

ES	Activity	EF
LS	Duration	LF

اور آخری دہائی EF سے LF ہوگا اور پہلے دہائی ES اور EF ہوگا
 اور duration دو تہائی ہوگی -
 Note:- 13
 پہلے EF دوسرے ES ہوگا
 جب forward کریں گے تو پہلی دہائی اور تیسری دہائی اور (-) کریں گے
 اور جب Reverse کریں گے تو تیسری دہائی اور پہلی دہائی اور (-) کریں گے



Total Slack:-

It have 2 formulas

$$TS = LF - EF \quad \text{or} \quad TS = LS - ES$$

$$A = 10 - 10 = 0$$

$$B = 17 - 7 = 10$$

$$C = 17 - 15 = 2$$

$$D = 23 - 23 = 0$$

$$E = 21 - 19 = 2$$

$$F = 35 - 35 = 0$$

$$G = 35 - 33 = 2$$

c) Critical Path:-

Critical Path is a path in which total slack value is 0

So,

$$CP \Rightarrow A, D, F$$

To find Critical Path value we can add the time of its activity

$$CPV = A + D + F$$

$$= 10 + 13 + 12$$

$$= 35 \text{ Ans}$$

b)

For Non-Critical Path:-

These are two ways for

Non-CP.

1) A, C, E, G

$$\begin{aligned} \text{Length/ C.P.V} &= A + C + E + G \\ &= 10 + 5 + 4 + 14 \\ &= 33 \end{aligned}$$

2) B, E, G

$$\begin{aligned} \text{C.P.V} &= B + E + G \\ &= 7 + 4 + 14 \\ &= 25 \end{aligned}$$

Q (2015)

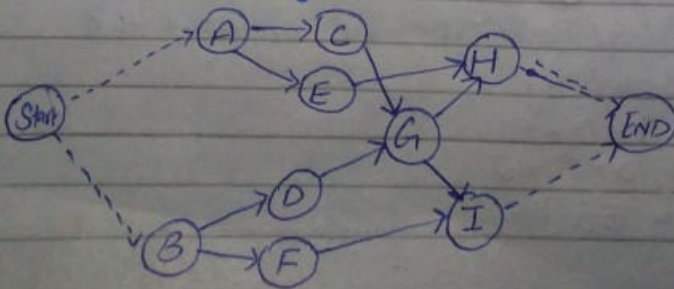
Activity

Time

Pre-decessor

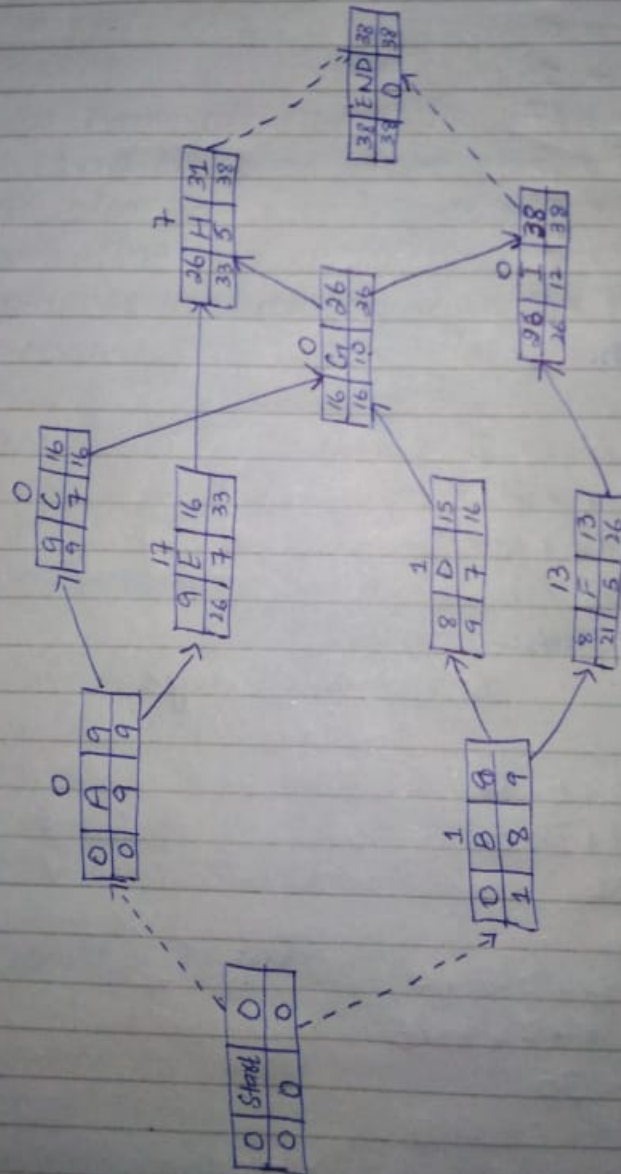
A	9	—
B	8	—
C	7	A
D	7	B
E	7	A
F	5	B
G	10	C, D
H	5	E, G
I	12	G, F

Ans: a) Network Diagram:-



b)

ES	Activity	EF
LS	Time	LF



Total Slack:- $TS = \text{LF} - \text{EF}$

$$A = 9 - 9 = 0$$

$$B = 9 - 8 = 1$$

$$C = 16 - 16 = 0$$

$$D = 16 - 15 = 1$$

$$E = 33 - 16 = 17$$

$$F = 26 - 13 = 13$$

$$G = 26 - 26 = 0$$

$$H = 38 - 31 = 7$$

$$I = 38 - 38 = 0$$

c) Critical Path:-

$$C.P. \rightarrow A, C, G, I$$

$$\begin{aligned} C.P.V &= A + C + G + I \\ &= 9 + 7 + 10 + 12 \\ &= 38 \text{ Ans.} \end{aligned}$$

Non-Critical Path:-

It has many ways:-

1) A, E, H

$$\begin{aligned} N.C.P.V &= A + E + H \\ &= 9 + 7 + 5 \\ &= 21 \end{aligned}$$

2) B, D, G, H

$$\begin{aligned} N.C.P.V &= B + D + G + H \\ &= 8 + 7 + 10 + 5 \\ &= 30 \end{aligned}$$

3) B, F, H

$$\begin{aligned} N.C.P.V &= B + F + H \\ &= 8 + 5 + 5 = 18 \end{aligned}$$

Q Define Software Configuration Management?
Explain its processes?
(2013), (2017)

Ans:- SCM:-

Software Configuration Management (SCM) is a set of management disciplines within a software engineering process to develop a baseline. SCM encompasses the disciplines and techniques of initiating, evaluating and controlling change to software products during and after a software project. It is an umbrella activity that is applied throughout the software process because change can occur anytime. SCM is an integral part of Software Quality Assurance (SQA).

SCM Processes:-

The Various SCM processes are:-

- 1) Identifying Objects
- 2) Controlling Versions
- 3) Controlling Changes
- 4) Auditing
- 5) Communication Changes.

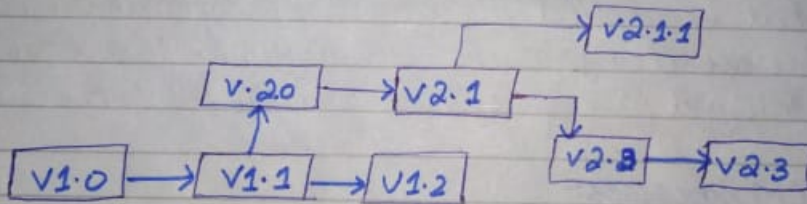
1) Identifying Objects:-

The first activity in SCM involves software configuration items (SCI's). SCI is an aggregation of software that is designed for configuration managements. It is

treated as a single entity in the configuration management process. e.g.: design a document, program code, test cases etc.

2) Controlling Versions:-

Version control combines procedures and tools to manage different versions of configuration objects that are created during software product development. A new version is defined when major changes have been made to one or more objects.



3) Controlling Changes:-

Change Control combines human procedures and automated tools to provide a mechanism for controlling change. The purpose of control change is to monitor and control changes in order to baseline SCI's. There are various reasons that trigger changes:-

- Suggestions and ideas from brainstorming sessions
- Feedback from clients
- Problem report might call for a change
- Addition to functionality.

4) Auditing:-

Configuration Audit is conducted formally by the SGA group in projects where SCM is a formal activity. It is ensure that changes has been properly implemented. The Identification, version control and change control tasks helps the developer to maintain order in an environment of change.

5) Communication changes:-

The last process of SCM is communication changes. This task ensures communication changes b/w different members in the project. The task is all about status reporting. In large projects, there is a possibility of miscommunication among various people involved in the project. Status Reporting include following questions:-

- What happened?
- When did it happen?
- Who did it?
- What else will be affected?

Q PV = Rs 10,000, EV = Rs 5000, AC = Rs 15,000
(2015)

PV = Planned Value

CV = Cost Variance

EV = Earned Value

SV = Schedule Variance

AC = Actual Cost

SPI = Schedule Performance Index

CPI = Cost Performance Index.

Formula:- $CV = EV - AC$, $SV = EV - PV$
 $CPI = EV/AC$, $SPI = EV/PV$

a) Calculate SV, CPI, SPI?

Sol:- $SV = EV - PV = 5000 - 10,000 = -5000$
 $CPI = EV/AC = 5000/15000 = 0.3333$
 $SPI = EV/PV = 5000/10,000 = 0.5$

b) Use SPI to estimate how is project doing, Is it head of schedule or behind the schedule?

Sol:- As,

$$SPI < 1$$

Therefore,

Project is behind the schedule.

Q what is Difference b/w Brainstorming and the Delphi technique for Risk Identification? Write some Advantages & Disadvantages for each approach?

2016

Index

Ans:- Brainstorming:-

Brainstorming is a technique by which a group attempts to generate ideas or find a solution for a specific problem by amassing ideas ~~the~~ spontaneously without judgement. This approach can help the group create a comprehensive list of risks to address later in the qualitative and quantitative risk analysis. Brainstorming can be used in businesses to produce greater numbers of ideas.

Advantages:-

- group members can bond over ideas
- Ease of use.
- Identify list of ideas in a short period of time.
- Fast pace, even energizing activity.

Dis Advantages:-

- longer problem resolution
- No mechanism for converging ideas
- Wasting of group's time on ideas that are wildly impractical.

Delphi technique:-

Delphi technique is to derive a consensus among a panel of experts who make predictions about future developments. The Delphi technique is a systematic, interactive forecasting procedures based on independent and anonymous input regarding future events. This technique used to improve group decisions. This technique uses repeated rounds of questioning, written responses, feed-back to earlier-round responses to take advantage of group input.

Advantages:-

- Opportunities for large numbers of people.
- Focus is on ideas rather than individuals.
- Allows for identification of priorities.
- Many experts involved to work quickly.

Dis-Advantages:-

- Complexity of data analysis.
- Large amount of time to conduct several rounds.
- Time Consuming
- Difficulty getting a panel of experts.

Q Explain any one technique to "Sequence the activities" by taking relevant example?
(2018)

Ans:- Sequence Activities:-

Sequence Activities is the process of identifying and documenting relationships among the project activities. The key benefit of this process is that it defines the logical sequence of work to obtain the greatest efficiency given all project constraints.

Tools and Techniques:-

The tools and techniques that are used in Sequence Activities are:-

1) Precedence Diagramming Method:-

PDM is a technique in which activities are linked, by one or more logical relationship, to show the sequence in which activities are to be performed. It uses four types of predecessor relationships.

- Early Start (ES)
- Early Finish (EF)
- Late Start (LS)
- Late Finish (LF)

2) Mandatory & Discretionary dependencies:-

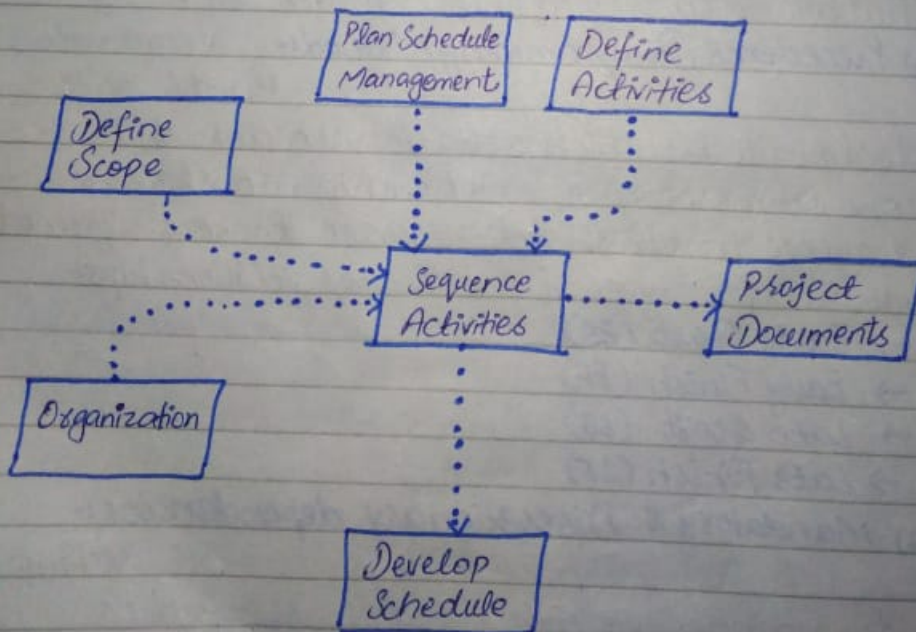
Mandatory dependencies are either legally required or simply a necessary part of the work. Discretionary dependencies are when you have one resource

doing several activities. Discretionary dependencies can be changed but mandatory dependencies can not be changed.

3) Lags and Leads:-

Lags and Leads are ways to put time between two activities in order to accurately define the logical relationship. In lags, the amount of time of a activity is added and in lead, the amount of time of a activity is subtracted.

Diagram of Sequence Activities:-



Q Define Slack Time? (2018)

Ans:- Slack time can be defined as the amount of time an task can be delayed without causing another task to be delayed or impacting the completion date of your project. Slack time is used to help people figure out just how much time is available between the various steps of a project. Slack time is created when certain tasks of the project take more time than others.

Formula:-

There are two formulas to find slack time:-

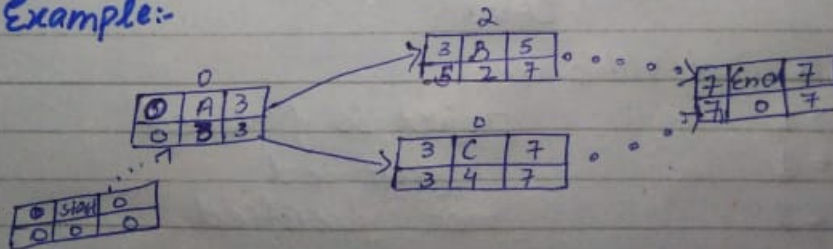
$$ST = \text{Late Finish} - \text{Early Finish}$$

$$ST = \text{Late Start} - \text{Early Start}$$

$$ST = \text{Late Start} - \text{Early Start}$$

Slack time is also known as Total Slack.

Example:-



$$\text{So, } A \Rightarrow ST = 3 - 3 = 0$$

$$B \Rightarrow ST = 7 - 5 = 2$$

$$C \Rightarrow ST = 7 - 7 = 0$$

Q Explain the Criteria to select project by using "cost benefit Analysis?"

(2018)

Ans:- The purpose of cost Benefit analysis in PM is to have a systematic approach to Figure out the pluses and minuses of various paths through a project, including transactions, tasks, business requirements and investments. The primary benefit of meeting quality requirements is less rework which means higher productivity, lower costs and increased stakeholders satisfaction. The process of cost benefit analysis are:-

- Project goals and objectives
- Note Alternatives
- List Stakeholders
- Determine the outcome of cost & Benefits
- Do analysis and making decisions.

Criteria:- Benefits - Disbenefits
Costs

Benefits / Cost ratio is:

> 1	project is justified
$= 1$	justified
< 1	not justified

Q1 What is a project?

(2016), (2017)

Ans:- A Set of related tasks which have a specific goal is called project. According to PMI (Project Management Institute), It is a temporary endeavor undertaken to create a unique product or service. Project may involve a single person or thousands. Project may be completed in hours, several months or years.

e.g:- Designing a new vehicle, Constructing a building etc.

Q3 What are the Project Management skills and competences for Project managers?

(2015), (2016)

Ans:- PM Skills:-

- Leadership
- Negotiating
- Team management
- Problem solving.
- Communication
- mentoring
- Risk Management

Competencies:-

- Execution
- Decision Making
- Critical thinking.
- Strategic Development
- Technical Competence

Ans:- 1) Project Administrator
2) Project Manager

Q5 Name the Knowledge areas of Project Management?
(2018)

Ans:- PMI has suggested 9 Knowledge areas:-

- 1) Project Integration Management.
- 2) " Communication "
- 3) " Time "
- 4) " Cost "
- 5) " Risk "
- 6) " Quality "
- 7) " Human Resource "
- 8) " Procurement "
- 9) " Scope "

Q9 Explain Project Monitoring and Controlling?
(2014)

Ans:- The Purpose of Project Monitoring and Controlling is to keep the team and management up to date on the project's progress. If the project deviates from the plan, then the Project managers take action to correct the problem. Monitoring & Controlling is done throughout the life of Project.

Q10 What is Project life Cycle and its phases?
(2015), (2018)

Ans:- Project life Cycle:-

A Project life Cycle is a collection of Project phases that defines

- what work will be performed in each phase
- what deliverables will be produced and when?
- who is involved in each phase, and
- how management will control and approve work produced in each phase.

Project life Cycle Phases:-

The Phases of Project life Cycle are:-

1) Initiation Phase:-

Initiation phase defines those processes that are required to start a new project. The purpose of this phase is to

determine what the project should accomplish. This phase mainly composed of two main activities 1) Develop a project Charter and 2) Identify Stakeholders. All the information related to the project are entered in both activities.

2) Planning Phase:-

Planning phase covers about 50% of the whole process. This phase determine the scope of the project as well as the objective of the project. It begins with the outputs of the initiation phase. The output of the planning phase serves as the input for the execution phase.

3) Execution phase:-

The Execution Phase consist of those activities that are defined in project management plan. This process involves managing stakeholder expectations, coordinating with people and resources, as well as performing other activities related to project deliverables.

4) Monitoring & Controlling phase:-

After Execution phase, to check the project is on right track, monitoring and controlling phase is active.

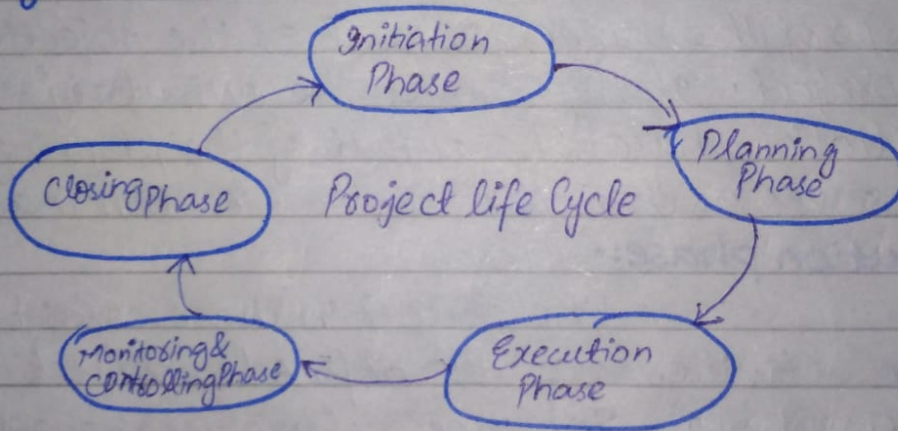
During this phase various changes and reviews to enhance the project performance is done.

This phase involves tracking, reviewing and regulating the progress in order to meet the objective of the project.

5) Closing Phase:-

Closing phase is the process that performs a controlled shutdown of the project at the end. In a project, there are three closure activities that are going on 1) Closure of the Product, 2) Closure of the project, 3) Closure of the resource behind the project.

Diagram:-



Q13 What are the processes involved in Project Scope Management?

(2016)

Ans:- There are five processes involved in Project Scope Management:-

1) **Scope Planning:-**

Deciding how the scope will be defined, verified and controlled.

2) **Scope Definition:-**

Reviewing the project charter and preliminary scope statement and adding more information.

3) **Creating the WBS:-**

Subdividing the major project deliverables into smaller, more management components.

4) **Scope Verification:-**

Formalizing acceptance of the project scope by key project stakeholders.

5) **Scope Control:-**

Controlling changes to project scope which impact project cost and time goals.

Q14 What is WBS and WBS Dictionary?
(2014), (2016)

Ans:- WBS:-

A WBS is a deliverable-oriented grouping of the work involved in a project that defines the total scope of the project. A (Work Breakdown Structure) is a foundation document that provides the basis for planning and managing project schedules, costs, resources and changes.

WBS Dictionary:-

A WBS dictionary is a document that describes detailed information about each WBS item. The WBS and WBS dictionary form the ~~scope~~ scope baseline, which is used to measure performance in project ~~scope~~ goals.

Q22 Explain Qualitative and Quantitative Risk Analysis?

(2014), (2015)

Ans:- Qualitative Risk Analysis:-

A Qualitative Risk analysis prioritizes the identify project risks using a pre-defined rating scale. Risks will be scored based on their probability. Probability is commonly ranked on a zero to one scale and the impact scale is defined from one to five scale. A Qualitative Risk analysis will also include the appropriate risks, either source-based or effect based.

Tools & Techniques for Performing Qualitative Risk Analysis:-

- Risk Probability and Impact Assessment
- Risk Categorization
- Expert Judgement
- Probability and Impact Matrix.

Quantitative Risk Analysis:-

A Quantitative Risk analysis is a further analysis of the highest priority risks during the project. You will need a highly quality data, a well developed project model and prioritized list of risks usually from performing a qualitative risk analysis. It provides a quantitative approach by making decisions when there is uncertainty.

Tools & Techniques for Performing Quantitative Risk Analysis:-

- Data Gathering
- Expert Judgement
- Data Analysis (Simulations, Sensitivity analysis, Decision tree analysis)
- Interpersonal and team skills.

Difference Blw Both:-

Qualitative

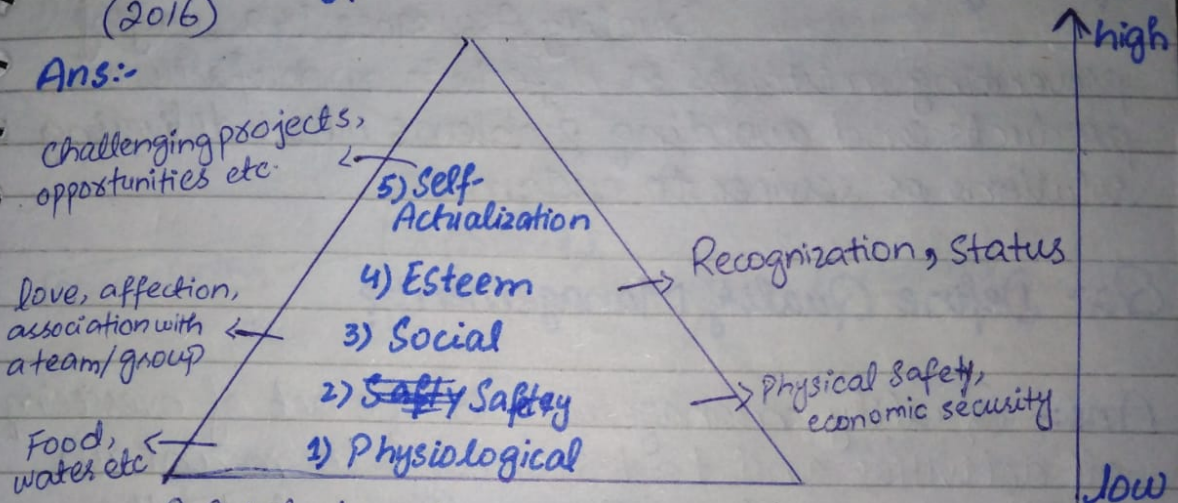
- Risk-level
- Quick and easy to perform
- No special tools or softwares required
- Qualitative Risk analysis is a probability Rank.
- It is applied in each possible risks.

Quantitative

- Project-level
- Slow and time consuming
- May required special tools
- Quantitative Risk analysis is a mathematical calculation.
- It is applied to only high-medium probability or impact risks.

Q25 What do you mean by hierarchy of needs theory?
(2016)

Ans:-



A Satisfied need is no longer a motivator

Maslow developed a hierarchy of needs that states that people's behaviours are guided or motivated by a sequence of needs.

Q26 Differentiate "Quality Control" and "Quality Assurance"?

(2015), (2018)

Ans:- **Quality Control:-**

Quality Control is defined as the process of setting standards and testing to make sure something, like a product or service, is done correctly.

Quality Assurance:-

Quality Assurance is a way of preventing mistakes or defects in manufactured products and avoiding problems when delivering solutions or services to customers.

Q29 Differentiate b/w Project life Cycle and Software development life Cycle?
(2017)

Ans:- SDLC:-

SDLC is a framework for describing the phases involved in developing and maintaining information system. It only applies to software projects.

Project Life Cycle:-

It applies to all projects regardless of the products being produced. It is not applies to software projects only.

Q30 Explain Quality with respect to triple Constraints?
(2017)

Ans:- Project Quality is affected by balancing these three factors.

$$Q = f(T, C, S)$$

where Q is Quality, S is Scope, C is Cost and T is Time.

Project fails when:

- a) Estimates are faulty
- b) Time, talent and resources are insufficient or incorrectly applied.